

LAB EXPERIMENT RECORD

Set 12: Website Traffic

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

AIM

To analyze website traffic data using a line chart, bar chart, table, and Tableau dashboard for interactive exploration of website traffic.

PROGRAM 1: LINE CHART OF DAILY PAGE VIEWS

Aim

To create a line chart to visualize the trend in daily page views over time.

R Code

R

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
```

```

page_views <- c(1500, 1600, 1400)

plot(date, page_views,

     type = "o",

     xlab = "Date",

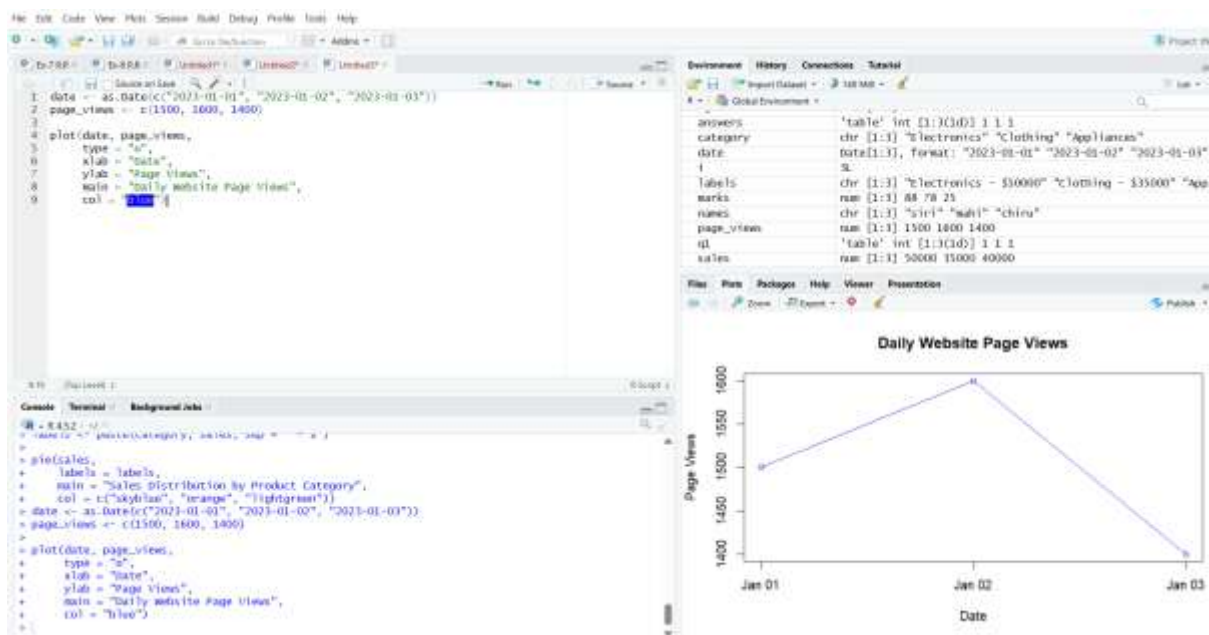
     ylab = "Page Views",

     main = "Daily Website Page Views",

     col = "blue")

```

Output



A line chart showing the daily page views from January 1 to January 3, 2023.

Observation

The page views increased from 1500 on January 1 to 1600 on January 2. On January 3, the page views decreased to 1400.

PROGRAM 2: BAR CHART OF TOP N CLICK-THROUGH RATES

Aim

To generate a bar chart showing the top days with the highest click-through rates.

R Code

R

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
```

```
ctr <- c(2.3, 2.7, 2.0)

website <- data.frame(date, ctr)

website <- website[order(-website$ctr), ]

top_n <- head(website, 3)

barplot(top_n$ctr,

        names.arg = top_n$date,

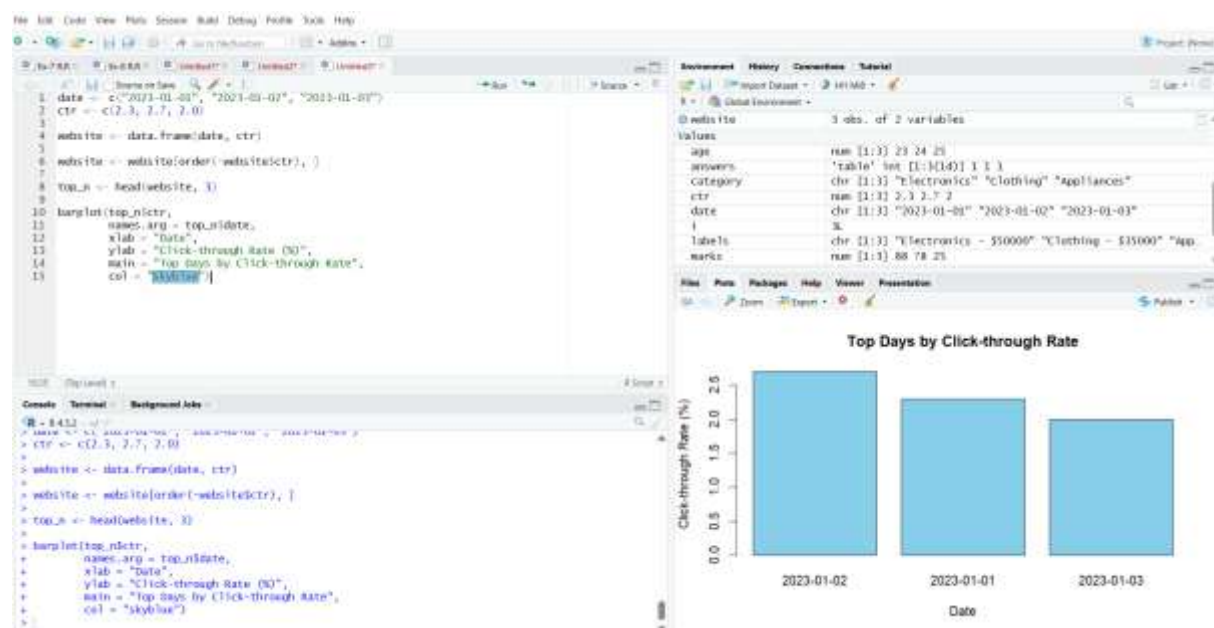
        xlab = "Date",

        ylab = "Click-through Rate (%)",

        main = "Top Days by Click-through Rate",

        col = "skyblue")
```

Output



A bar chart showing the click-through rate for the three days.

Observation

January 2 has the highest click-through rate of 2.7%, followed by January 1 with 2.3% and January 3 with 2.0%.

PROGRAM 3: TABLE OF WEBSITE TRAFFIC DATA

Aim

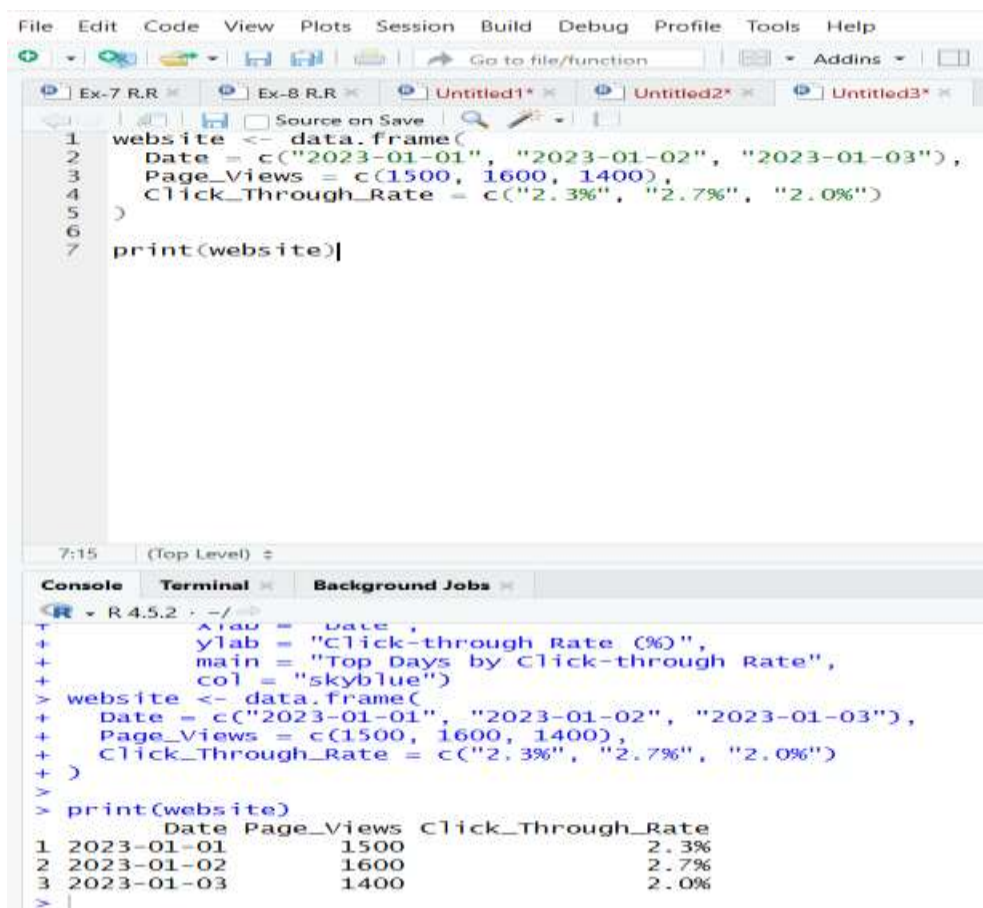
To build a table showing the website traffic data for each day.

R Code

R

```
website <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
  Page_Views = c(1500, 1600, 1400),  
  Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
)  
print(website)
```

Output



The screenshot shows the R Studio interface. The script editor at the top contains the R code to create a data frame and print it. The console at the bottom shows the execution of the code, including the creation of a plot (which is not displayed) and the printing of the data frame.

```
1 website <- data.frame(  
2   Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
3   Page_Views = c(1500, 1600, 1400),  
4   Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
5 )  
6  
7 print(website)
```

```
> website <- data.frame(  
+   Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
+   Page_Views = c(1500, 1600, 1400),  
+   Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
+ )  
> print(website)  
  Date Page_Views Click_Through_Rate  
1 2023-01-01      1500           2.3%  
2 2023-01-02      1600           2.7%  
3 2023-01-03      1400           2.0%
```

Observation

The table clearly displays the date, number of page views, and click-through rate for each day.

PROGRAM 4: TABLEAU DASHBOARD

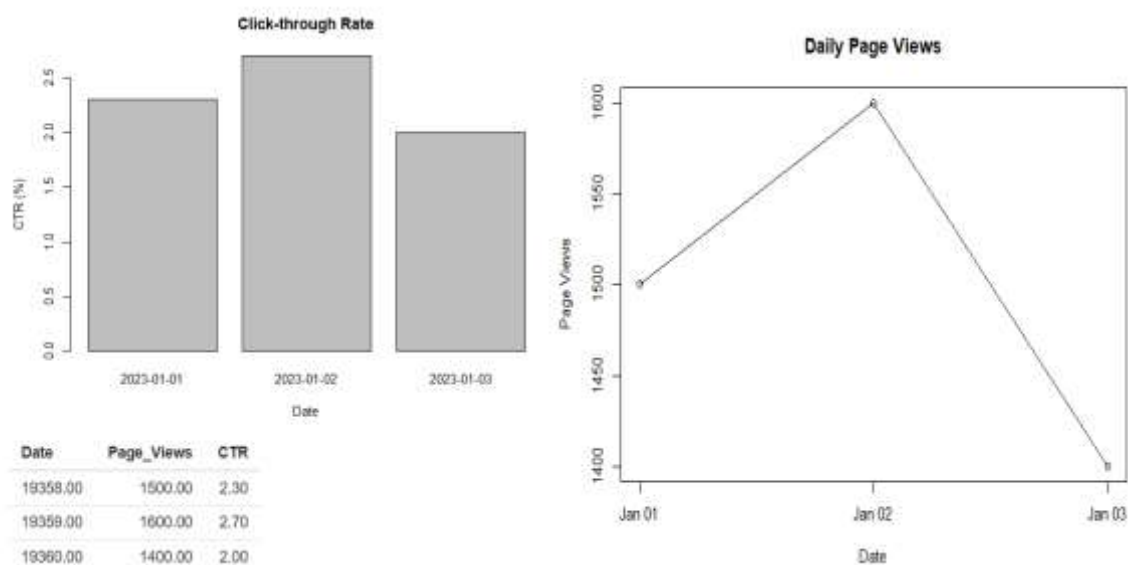
Aim

To develop a Tableau dashboard combining the line chart, bar chart, and table for interactive exploration of website traffic data.

Steps

1. Open Tableau.
2. Import the Website Traffic dataset.
3. Create a worksheet for the **Line Chart**.
4. Create another worksheet for the **Bar Chart**.
5. Create a third worksheet for the **Website Traffic Table**.
6. Select **Dashboard** → **New Dashboard**.
7. Drag the line chart into the dashboard.
8. Drag the bar chart into the dashboard.
9. Drag the website traffic table into the dashboard.
10. Arrange all three components properly.
11. Add suitable titles and labels.
12. Add filters if required for interactive exploration.
13. Save the Tableau dashboard.

Output



An interactive Tableau dashboard combining the line chart, bar chart, and website traffic table.

Observation

The Tableau dashboard provides a clear and interactive view of website traffic. It helps users compare daily page views and click-through rates while viewing the detailed data in table form.